

Idle Time Insertion for CSR [1/2]

- 마지막 stage의 machine 별로(m) 반복:

- $j \leftarrow |\mathcal{J}_m|$

- While $j > 0$:

- ① 해당 operation을 최좌측 operation으로 하는 block (idle time이 0인 operation set, S_M) 생성

- ② Δ_2 정의:

- Block의 오른쪽에 job이 있는 경우 해당 간격
 - 아닌 경우 ∞

- ③ S_M 의 subset 정의 (not partition)

- $S_E = \{j | KC'_j < d_j^-, j \in S_M\}$
(early now)

- $S_{E1} = \{K(C'_j + 1) \leq d_j^- \mid j \in S_E\}$
(full benefit by unit shift)

- $S_{E2} = \{K(C'_j + 1) > d_j^- \mid j \in S_E\}$
(partial benefit by unit shift)

- $S_D = \{j | d_j^- \leq KC_j, K(C'_j + 1) \leq d_j^+, j \in S_M\}$
(on-time now, on-time after unit shift)

- $S_T = \{j | d_j^+ < K(C'_j + 1), j \in S_M\}$
(tardy after unit shift)

- $S_{T1} = \{d_j^+ \leq KC'_j \mid j \in S_T\}$
(full penalty by unit shift)

- $S_{E2} = \{d_j^+ > KC'_j \mid j \in S_T\}$
(partial penalty by unit shift)

- 한 operation이 S_E 와 S_T 에 동시에 속할 수 있으나,
($S_E \cup S_T$) $\cup S_D = \emptyset$

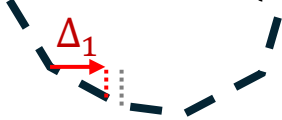
Idle Time Insertion for CSR [2/2]

④ $\sum_{k \in S_{E1}} w_j^- K + \sum_{k \in S_{E2}} w_j^- (d_j^- + K C'_j) > \sum_{k \in S_{T1}} w_j^+ K + \sum_{k \in S_{T2}} w_j^+ (K(C'_j + 1) - d_j^+)$
 (If right-shifting by K reduces obj. func.)

▪ True: right-shift

• If $S_{E2} \cup S_{T2} \neq \emptyset$, $\Delta_1 := K$

• Else, $\Delta_1 := \min \left\{ \min_{k \in S_E} \left(\left\lfloor \frac{d_k^-}{K} \right\rfloor - C'_k \right), \min_{k \in S_D} \left\{ \left\lfloor \frac{d_k^+}{K} \right\rfloor - C'_k \right\} \right\}$



• $\Delta := \min\{\Delta_1, \Delta_2\}$

• If $\Delta > 0$, $C'_k \leftarrow C'_k + \Delta \forall k \in S_M$ (right-shift);
 If $\Delta = 0$, $j \leftarrow j - 1$

▪ False: go to the next(left) job

• $j \leftarrow j - 1$